

Lorenzo Minervino

Software Engineer | Backend & Distributed Systems

Argentina · lminervino18@gmail.com · +54 9 299 577 8365

linkedin.com/in/lminervino18 · github.com/lminervino18 · lminervino18.com

PROFESSIONAL SUMMARY

Computer Engineering candidate at the University of Buenos Aires with all coursework completed and thesis defense pending. Focused on backend development and distributed systems, with hands-on experience in Rust, Java and Python across academic and personal projects covering replication, concurrency, fault tolerance and REST services. Currently working as an AI Trainer, evaluating technical responses related to programming and mathematics.

TECHNICAL SKILLS

Programming Languages: Java, Python, Rust, C, JavaScript/TypeScript

Backend: REST APIs, Spring Boot, Node.js (NestJS), Microservices, JWT Authentication

Databases: MySQL, PostgreSQL, MongoDB

Distributed Systems: Replication, Consistency Models, Fault Tolerance, Leader Election, Replicated Logs, Message Queues (RabbitMQ)

Systems: Concurrency, TCP Socket Programming, Operating Systems, Linux

Tools: Git, Docker, Docker Compose, Agile (Scrum, Kanban)

SELECTED PROJECTS

Oxidized Neural Orchestra — Final Degree Project · Rust, Python (PyO3), Docker · team of 3

github.com/lminervino18/oxidized-neural-orchestra

- Distributed system that trains neural networks across a cluster of nodes, implementing three strategies: parameter server, ring all-reduce, and a hybrid that promotes workers into parameter servers once gradients converge.
- Structured as a Cargo workspace with a custom TCP communication layer, a role-agnostic node binary assigned its role at runtime by an orchestrator, Python bindings through PyO3, a terminal UI for live monitoring, and Docker Compose to simulate multi-node clusters.

Rustic Airlines — Rust, Docker · team of 4 · github.com/lminervino18/rustic-airlines

- Distributed flight management system backed by a Cassandra-inspired database engine, supporting multi-node replication, configurable consistency levels, fault tolerance and cluster growth through seed-node discovery, with a driver layer and a flight simulator on top.

Distributed Transaction Analysis Pipeline — Python, RabbitMQ, Docker · team of 3

github.com/lminervino18/distribuidos-tp-money-laundering-analysis

- Multi-stage pipeline (filter, join, group-by, aggregate) distributed over RabbitMQ queues, with replicated stages coordinated by a ring protocol for end-of-stream detection.
- Fault tolerance built on leader/replica supervisors with heartbeats, automatic node restarts and checkpointing, validated through a chaos harness that kills and restores random nodes.

CONICET Researchers Platform — Java, Spring Boot, Spring Data JPA, MySQL, React · team of 2

github.com/lminervino18/researchers-page-conicet

- REST API for publishing scientific work, team information and outreach content, with JWT-based authentication and a React/TypeScript frontend consuming the endpoints.

PROFESSIONAL EXPERIENCE

AI Trainer — Outlier · Remote · Jan 2025 – Present

- Reviews and corrects AI-generated answers on programming and mathematics, assessing technical correctness, reasoning and clarity of the explanations.
- Writes structured feedback that documents each error and the reasoning behind the correction.

EDUCATION

Computer Engineering — University of Buenos Aires (UBA)

All coursework completed. Thesis defense pending — Expected 2026.

LANGUAGES

Spanish: Native · English: Upper-Intermediate (B2), Cambridge First Certificate (FCE)